Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam C · Form 1: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. Add identical small masses to both sides of an initially unbalanced Atwood machine. How does acceleration magnitude change?

(A) It stays fixed.

(B) Only its direction changes.

(C) It increases.

(D) It decreases.

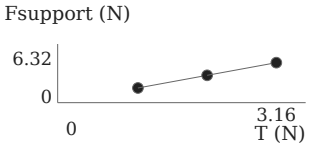
2. A negligible-weight fixed pulley has two vertical rope segments below it. A support holds its axle. T (N) -> Fsupport (N): (1.16, 2.32); (2.16, 4.32); (3.16, 6.32). By what factor does the first-to-third change exceed the first-to-second change?

(A) 1.2

(B) 2.8

(C) 2

(D) 3.6



3. A heavier Atwood mass is moving upward immediately before reversal. Which quantity changes sign at reversal while the rope remains taut?

(A) The mass difference

(B) Velocity of the heavy mass

(C) Its acceleration

(D) Tension magnitude

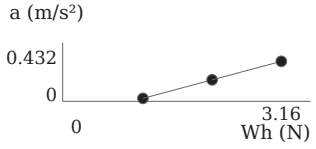
4. A horizontal cart slides toward an ideal pulley. Cart plus hanger mass is fixed at 5 kg; a constant 1 N resistance acts on the cart. Use g=10 m/s². Wh (N) -> a (m/s²): (1.16, 0.032); (2.16, 0.232); (3.16, 0.432). Which horizontal setting produces a reading of 0.132?

(A) 1.66

(B) 0.996

(C) 2.988

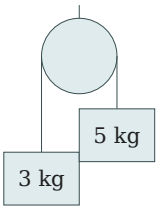
(D) 2.324



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. The right mass of an ideal Atwood machine is initially moving upward. Left mass is 3 kg, right mass is 5 kg. Use g=10 m/s².

a. Draw separate velocity and acceleration arrows for the right mass. **(1 pt)**



b. If initial upward speed is 1.0 m/s, find reversal time and tension at reversal. **(3 pt)**

6. A stationary incline block with 3 N of downhill weight is connected to a hanger over an ideal pulley. Static friction is sufficient; count uphill friction positive. Wh (N) -> fs signed (N): (1.16, 1.84); (2.16, 0.84); (3.16, -0.16).

a. Build the model for static friction on a ramp from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. Masses H and L form an Atwood pair with H>L. Move a small mass δ from the light side to the heavy side. Derive the change in acceleration, assuming 0<δ<L. **(3 pt)**